

Multiagent Fuzzy Reinforcement Learning With LLM for Cooperative Navigation of Endovascular Robotics

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Abstract—Endovascular interventions require precise, cooperative control of multiple instruments, such as guidewires and catheters, to navigate complex vascular anatomies. Current robotic systems, reliant on leader–follower control, depend heavily on operator expertise and lack intelligence. Learning-based methods, often limited to single-instrument control, fall short in complex clinical scenarios requiring multi-instrument coordination. This study proposes a multiagent fuzzy reinforcement learning (MAFRL) framework, guided by large language models (LLMs), for task-level autonomous, cooperative navigation in endovascular robotics. LLMs provide procedural priors and context-aware policy guidance, enabling adaptive decision-making for collaborative guidewire and catheter agents. Central to the framework, fuzzy reinforcement learning mitigates LLM-induced uncertainties by adaptively embedding clinical constraints into reward functions, ensuring strict adherence to procedural safety and precise alignment with the complexities of real-world endovascular interventions. Validated in a 3-D vascular simulation, this approach achieves superior navigation performance and procedural efficiency compared to conventional methods, underscoring the transformative potential of fuzzy reinforcement learning in advancing LLM-guided multiagent reinforcement learning for endovascular robotics.

Index Terms—Autonomous navigation, endovascular procedure, fuzzy reinforcement learning (FRL), large language models (LLMs), learning-based control.

I. INTRODUCTION

ENDOASCULAR interventions necessitate the precise coordination of multiple instruments [1], including guidewires, catheters, and stents [2], to navigate complex vascular structures and access target lesions for cardiovascular diagnosis and treatment [3]. The procedure, depicted in Fig. 1, typically begins with a small incision, followed by a percutaneous puncture of the femoral artery to establish an access channel. Subsequently, a guidewire is introduced and meticulously maneuvered through the vasculature to reach the lesion site. The catheter then advances along the guidewire to the target region, and the guidewire is withdrawn to enable the deployment of therapeutic agents or medical devices [4]. This process demands continuous real-time coordination and control of each instrument to ensure precise navigation and stable positioning, shown in Fig. 2, highlighting the technical expertise required of interventional cardiologists [5]. Despite the advantages of endovascular techniques, such as reduced complications, shorter recovery times, and lower postoperative pain compared to open surgery, their success remains contingent on the operator's proficiency in managing the dynamic interactions of the instruments, ultimately influencing procedural efficiency and patient outcomes [6].

Robot-assisted technologies have gained prominence as a promising solution for improving precision and safety in endovascular interventions [7]. However, current robotic systems primarily rely on a leader–follower paradigm. In this model, the robotic manipulator simply mirrors the operator's real-time movements. This approach lacks autonomous decision-making capabilities, limiting its potential for intelligent intervention. Although some systems integrate rudimentary intelligent control strategies, these depend on predefined rules or explicitly formulated control laws [8], [9], limiting their capacity to accommodate anatomical diversity. As a result, these systems struggle to dynamically address vessel deformation, patient-specific anatomical variations, and unforeseen obstacles [10], lacking the flexibility required for complex scenarios [11], [12]. Consequently, despite progress in robotic assistance, these systems remain heavily reliant on operator expertise, presenting

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significant barriers to achieving truly intelligent endovascular interventions.

Learning-based approaches have advanced robotic endovascular interventions by enabling data-driven control strategies that reduce reliance on preprogrammed paths [13], [14]. Reinforcement learning (RL) supports autonomous navigation by optimizing performance through pretraining in simulated environments [15], [16]. This adaptability suits high-precision, dynamic procedures [17]. However, existing RL methods typically focus on single-instrument tasks, such as guidewire navigation in Sim4EndoR [13] or catheter control in CathSim [18], limiting their applicability to multi-instrument clinical scenarios. Real-world endovascular interventions require coordinated control of guidewires and catheters to ensure navigational efficiency and safety, revealing a gap in current methodologies.

Large language models (LLMs) offer advanced reasoning and contextual knowledge integration, supporting the establishment of multiagent systems for endovascular interventions. This study proposes a multiagent fuzzy reinforcement learning (MAFRL) framework, guided by LLMs, for task-level autonomous, cooperative navigation in endovascular robotics. Fuzzy reinforcement learning (FRL) is integral to the framework, employing fuzzy systems to design adaptive reward functions that address uncertainties in LLM-generated policy guidance [19]. By incorporating clinical constraints into these reward functions, FRL ensures accurate, safe, and protocol-compliant decision-making, aligning with endovascular operational requirements [20]. LLMs provide procedural knowledge and context-aware policies, enabling cooperative agents—representing guidewires and catheters—to navigate patient-specific anatomies and complex vascular environments [21]. The integration of FRL enhances multiagent coordination, supporting efficient and safe navigation.

This study presents a framework for robot-assisted endovascular intervention based on MAFRL, with FRL as its core component. The work provides three main contributions: 1) a multiagent model representing guidewires and catheters as cooperative agents, achieving accurate navigation and delivery in complex vascular networks through LLM-guided strategies; 2) a customized RL framework with tailored state and action spaces, where FRL-designed reward functions, combined with LLM-driven policy guidance, enforce clinical constraints to ensure safe and adaptive navigation; and 3) validation in 3-D vascular models, demonstrating enhanced navigation accuracy, procedural efficiency, and robustness compared to existing methods.

II. RELATED WORKS

A. Learning-Based Navigation for Robotic-Assisted Endovascular Procedures

Recent advances in robotic-assisted endovascular interventions increasingly leverage learning-based techniques to enhance navigation autonomy and adaptability in complex vascular environments [22]. These methods, rooted in RL and optimization, improve decision-making for intelligent guidewire and catheter control. Robertshaw et al. [15] devised a method to infer reward functions from expert demonstrations, optimizing

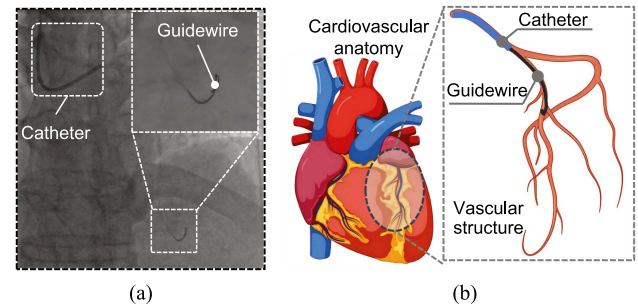


Fig. 1. Overview of catheter and guidewire intraoperative navigation in endovascular intervention. (a) DSA image captured during the endovascular procedure, showcases the coordinated manipulation of both the catheter and guidewire within the patient's vascular structure. (b) Schematic illustration of cardiovascular anatomy, emphasizing the interventional pathways and interactions between the catheter and guidewire during an endovascular intervention.

navigation policies for complex vascular networks. This approach reduces procedure time and operator radiation exposure, especially in urgent cases. Robertshaw et al. [10] utilized the Simulation Open Framework Architecture (SOFA) to create controlled simulation environments, enabling rapid testing and refinement of navigation policies under varied conditions for improved robustness. Li et al. [23] proposed the conservative actor-critic with smooth gradient (CASOG) algorithm, a RL-based method to address distribution shift and overfitting in skill acquisition. While effective, its reliance on precollected datasets limits adaptability to real-time, patient-specific scenarios. Similarly, Mei et al. [24] developed artificial intelligence (AI)-driven agents for vascular decision-making using 2-D digital subtraction angiography (DSA) images, but the lack of 3-D spatial data hinders precision in complex anatomies. Scarponi et al. [25] advanced autonomous catheter navigation with a deep RL framework, enhancing generalization to deforming anatomies via optimized training and motion estimation from single-view fluoroscopy. In simulation validation, it achieved a 95% success rate for randomized targets, showing promise for dynamic settings.

Current research largely focuses on single-instrument navigation (guidewires or catheters), yet clinical endovascular procedures require coordinated manipulation of both. This cooperative dynamic is critical for complex interventions. Future simulation environments must model guidewire-catheter interactions accurately, developing control strategies that address mechanical interdependencies, sequential patterns, and variable tissue interactions observed in real procedures.

B. Multiagent Reinforcement Learning (MARL) for Cooperative Task Execution in Medical Applications

MARL is a pivotal AI paradigm for cooperative task execution, enabling multiple agents to learn and coordinate in dynamic environments to achieve shared goals [26]. Leveraging decentralized policies and shared rewards, MARL enhances collaboration in complex decision-making tasks. In robotics, it extends beyond single-agent systems [27], with applications in unmanned aerial vehicle (UAV) coordination [28] and mobile

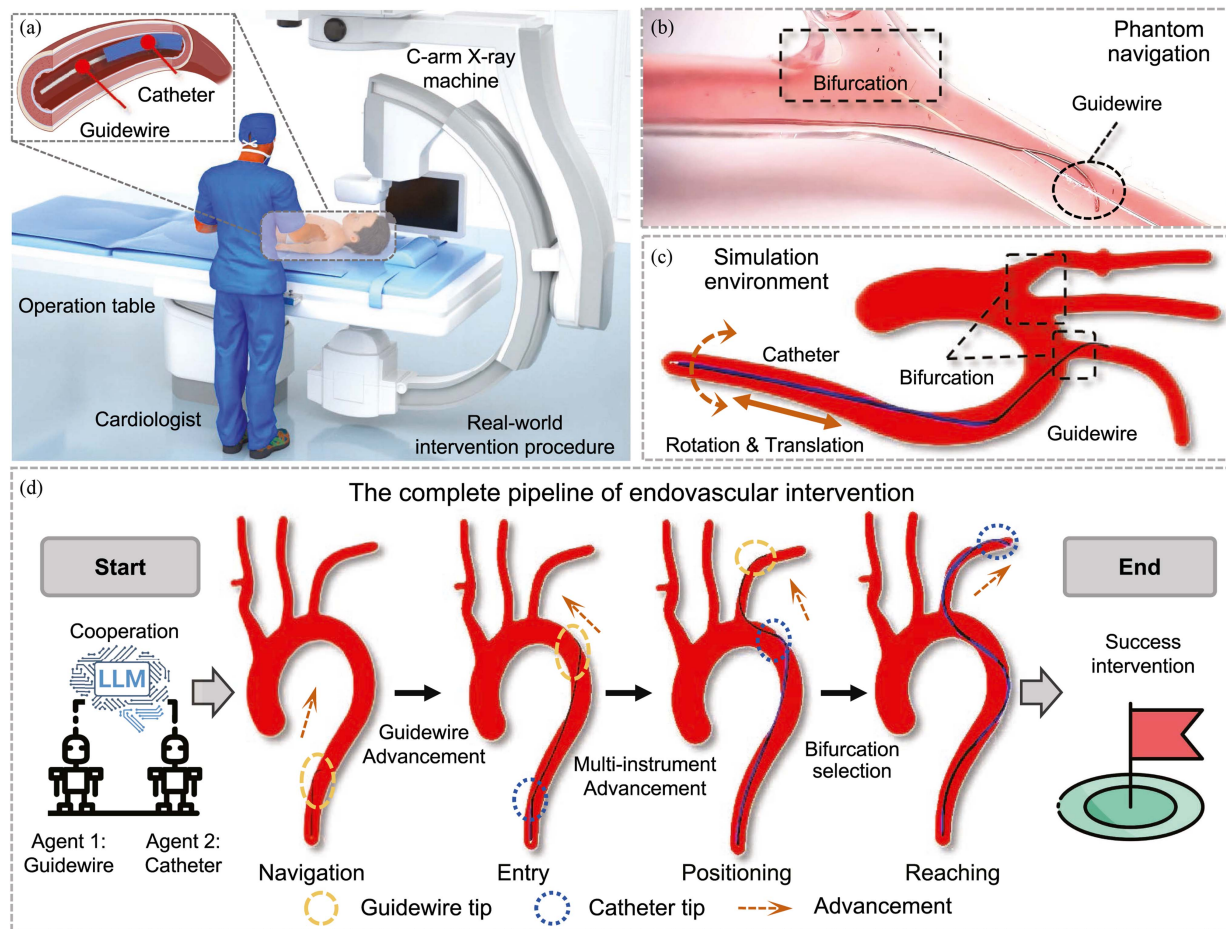


Fig. 2. Framework for robot-assisted endovascular intervention: Real-world, virtual, and pipeline perspectives. (a) Real-world operating room scenario, where a cardiologist manipulates a guidewire and catheter under C-arm X-ray guidance to perform endovascular intervention. (b) Simulation environment was developed for this study, enabling robotic-assisted autonomous navigation of multi-instruments to support endovascular intervention. Algorithms for agents' autonomy are constructed and evaluated within this virtual setting. (c) Complete procedural pipeline, spanning navigation (P1), entry (P2), positioning (P3), and reaching (P4), executed by cooperative agents: Agent 1 (guidewire) and Agent 2 (catheter).

robots [29], improving efficiency through synchronized control. While widely studied in robotics, MARL is nascent in robotic-assisted endovascular procedures, where it could enhance multi-instrument coordination within complex vascular structures, improving real-time decision-making and procedural efficiency.

The da Vinci Research Kit, equipped with multiple robotic arms including the endoscopic camera manipulator and patient-side manipulators, has advanced MARL in surgical robotics [30], [31], [32]. Munawar and Fischer [33] developed an asynchronous MARL framework with distributed control, suitable for laparoscopic tasks but less optimized for endovascular navigation requiring precise flexible instrument control. Scheikl et al. [34] proposed a MARL system for robot-assisted surgery, achieving a 71.2% reduction in procedure time and 98.0% fewer collisions compared to human teams, though tailored to laparoscopic applications. For endovascular interventions, Robertshaw et al. [15] applied inverse RL and soft actor-critic for autonomous guidewire navigation, enhancing success rates and procedure times, but their approach focuses

on single-instrument control, leaving multi-instrument MARL for complex tasks unexplored.

FRL enhances MARL by employing fuzzy systems to design adaptive reward functions that address uncertainties and incorporate clinical constraints, making it well-suited for complex endovascular navigation [26], [27]. Despite its potential, FRL remains underutilized in endovascular robotics, where it could enable robust multi-instrument coordination. LLMs complement FRL by providing advanced contextual knowledge and procedural guidance, generating context-aware policies that adapt to patient-specific vascular anatomies and intraoperative variability [21]. This study proposes an MAFRL framework, where FRL and LLMs are central. FRL constructs adaptive reward functions to ensure safe, precise, and protocol-compliant cooperative navigation of guidewires and catheters, while LLMs enhance policy generation with real-time procedural insights [20]. This integrated FRL–LLM approach improves adaptability and robustness, overcoming the limitations of prior nonfuzzy and single-instrument MARL methods in endovascular robotics.

III. METHODOLOGY

A. Problem Formulation

Cooperative navigation of a guidewire and catheter through complex vascular pathways to a target lesion is a critical task in endovascular procedures. This process requires precise control of the guidewire to explore intricate vascular geometries, while the catheter follows to deliver therapeutic interventions. The clinical challenge lies in coordinating these instruments to ensure safe, efficient navigation within the constrained vascular environment, where uncertainties in anatomical variations and instrument dynamics pose significant obstacles. To address these uncertainties, this study formulates the problem as a cooperative multiagent Markov decision process (MDP) enhanced by FRL. FRL employs fuzzy systems to model clinical constraints and mitigate uncertainties in multiagent coordination, ensuring robust navigation policies.

The MDP is defined by the tuple $\mathcal{M} = (\mathcal{S}, \{\mathcal{A}_i\}_{i=1}^2, \mathcal{P}, \{\mathcal{R}_i\}_{i=1}^2, \gamma)$, where the following holds.

- 1) $\mathcal{S}$: The state space, capturing configurations of the agents and vascular environment.
- 2) $\mathcal{A}_i$: The action space of agent i , where $i \in \{1, 2\}$ represents the guidewire and catheter, respectively.
- 3) $\mathcal{P} : \mathcal{S} \times \mathcal{A}_1 \times \mathcal{A}_2 \rightarrow \mathcal{S}$: The state transition function, modeling system dynamics with inherent uncertainties.
- 4) $\mathcal{R}_i : \mathcal{S} \times \mathcal{A}_1 \times \mathcal{A}_2 \rightarrow \mathbb{R}$: The FRL-based reward function for agent i , incorporating fuzzy constraints to address clinical objectives and safety.
- 5) $\gamma \in [0, 1)$: The discount factor.

The objective is to derive policies $\pi_i : \mathcal{S} \rightarrow \mathcal{A}_i$ for each agent i , maximizing the expected cumulative reward while ensuring safe, efficient navigation to the target lesion. FRL enhances this process by embedding fuzzy systems to handle uncertainties in state transitions and LLM-guided policy generation, aligning with clinical protocols.

B. Multiagent Modeling With Fuzzy-Based Clinical Coordination

1) *Agents Definition*: The task involves two primary agents: Agent 1 (guidewire) and Agent 2 (catheter). Agent 1 navigates vascular pathways to the target lesion, adjusting position, orientation, and curvature. Agent 2 advances over the guidewire, maintaining proximity to deliver interventions. FRL ensures robust coordination by incorporating fuzzy systems to enforce clinical constraints, preventing the catheter from overtaking the guidewire, thus addressing uncertainties in instrument interactions and ensuring safety.

2) *State Space*: Let $\mathbf{p}_w = (x_w, y_w, z_w)$ and $\mathbf{p}_c = (x_c, y_c, z_c)$ denote the guidewire and catheter tip positions, respectively. The state space is defined as $\mathcal{S} = \mathcal{S}_w \times \mathcal{S}_c$, where

$$\mathcal{S}_w = \{\mathbf{p}_w, \theta_w, \kappa_w, d_{wg}\}, \quad \mathcal{S}_c = \{\mathbf{p}_c, v_c, l_{wc}, d_{wc}\}.$$

Here, $\theta_w \in [0, 2\pi)$ is the guidewire tip orientation, $\kappa_w \in [\kappa_{\min}, \kappa_{\max}]$ is the guidewire curvature, $d_{wg} = \|\mathbf{p}_w - \mathbf{p}_g\|$ is the distance to the target lesion at $\mathbf{p}_g$, $v_c \in [v_{c,\min}, v_{c,\max}]$ is the catheter insertion velocity, $l_{wc} = s_w - s_c$ is the arc-length

difference, and $d_{wc} = \|\mathbf{p}_w - \mathbf{p}_c\|$ is the distance between tips. FRL models uncertainties in these state variables using fuzzy sets, ensuring robust state representations for complex vascular environments.

3) *Action Space With Clinical Constraints*: The action spaces are

$$\mathcal{A}_w = \{\Delta v_w, \Delta \theta_w\}, \quad \mathcal{A}_c = \{\Delta v_c, \Delta \theta_c\}$$

where $\Delta v_w \in [\Delta v_{w,\min}, \Delta v_{w,\max}]$ adjusts guidewire translation velocity, $\Delta \theta_w \in [\Delta \theta_{w,\min}, \Delta \theta_{w,\max}]$ adjusts guidewire rotation, $\Delta v_c \in [\Delta v_{c,\min}, \Delta v_{c,\max}]$ adjusts catheter translation velocity, and $\Delta \theta_c \in [\Delta \theta_{c,\min}, \Delta \theta_{c,\max}]$ adjusts catheter rotation. FRL enforces clinical constraints using fuzzy systems, ensuring $\Delta v_c \leq \Delta v_w$ when $l_{wc} \leq l_{\max}$ and $d_{wc} \geq d_{\min}$. These fuzzy constraints mitigate uncertainties in action selection, preventing the catheter from overtaking the guidewire and maintaining safe distances to avoid complications.

C. FRL With LLM Integration

To address the coordination challenges in guidewire-catheter navigation, aMAFRL framework is developed for cooperative navigation in endovascular robotics, integrating LLMs for contextual guidance and FRL as the core mechanism to address uncertainties in LLM outputs, instrument dynamics, and vascular anatomies. The LLM embeddings are initialized with pretrained weights and continuously updated during training to adapt to real-time contexts. Context-aware policy generation refers to the process of producing navigation policies that dynamically adapt to real-time environmental conditions, such as vascular topology and instrument states, as well as procedural objectives, ensuring clinically relevant and safe actions. The framework employs a centralized training and decentralized execution paradigm, aligning with clinical practice where guidewires and catheters operate independently yet require coordinated control (see Fig. 3). FRL leverages a Takagi–Sugeno (T-S) fuzzy inference system to model clinical constraints, ensuring robust navigation policies under uncertainty.

In centralized training, each agent accesses the global state $\mathcal{S}$ and joint action tuple (a_1, a_2) . A centralized critic function $Q_i : \mathcal{S} \times \mathcal{A}_1 \times \mathcal{A}_2 \rightarrow \mathbb{R}$, implemented as a neural network with two fully connected layers and ReLU activations, evaluates action values

$$Q_i(\mathcal{S}, a_1, a_2; \theta_i^Q) = f_{\text{ReLU}}(W_2 \cdot f_{\text{ReLU}}(W_1 \cdot [\mathcal{S}, a_1, a_2] + b_1) + b_2)$$

where W_1, W_2, b_1 , and b_2 are learnable parameters, and θ_i^Q denotes the critic's weights. The critic incorporates T-S fuzzy constraints to mitigate uncertainties in state transitions and instrument dynamics. In decentralized execution, each agent operates on local observations $o_i \in \mathcal{O}_i$, executing transformer-based policies $\pi_i : \mathcal{O}_i \rightarrow \mathcal{A}_i$, initialized with LLM embeddings. The framework extends the multi-agent deep deterministic policy gradient (MADDPG) algorithm, with FRL designing adaptive

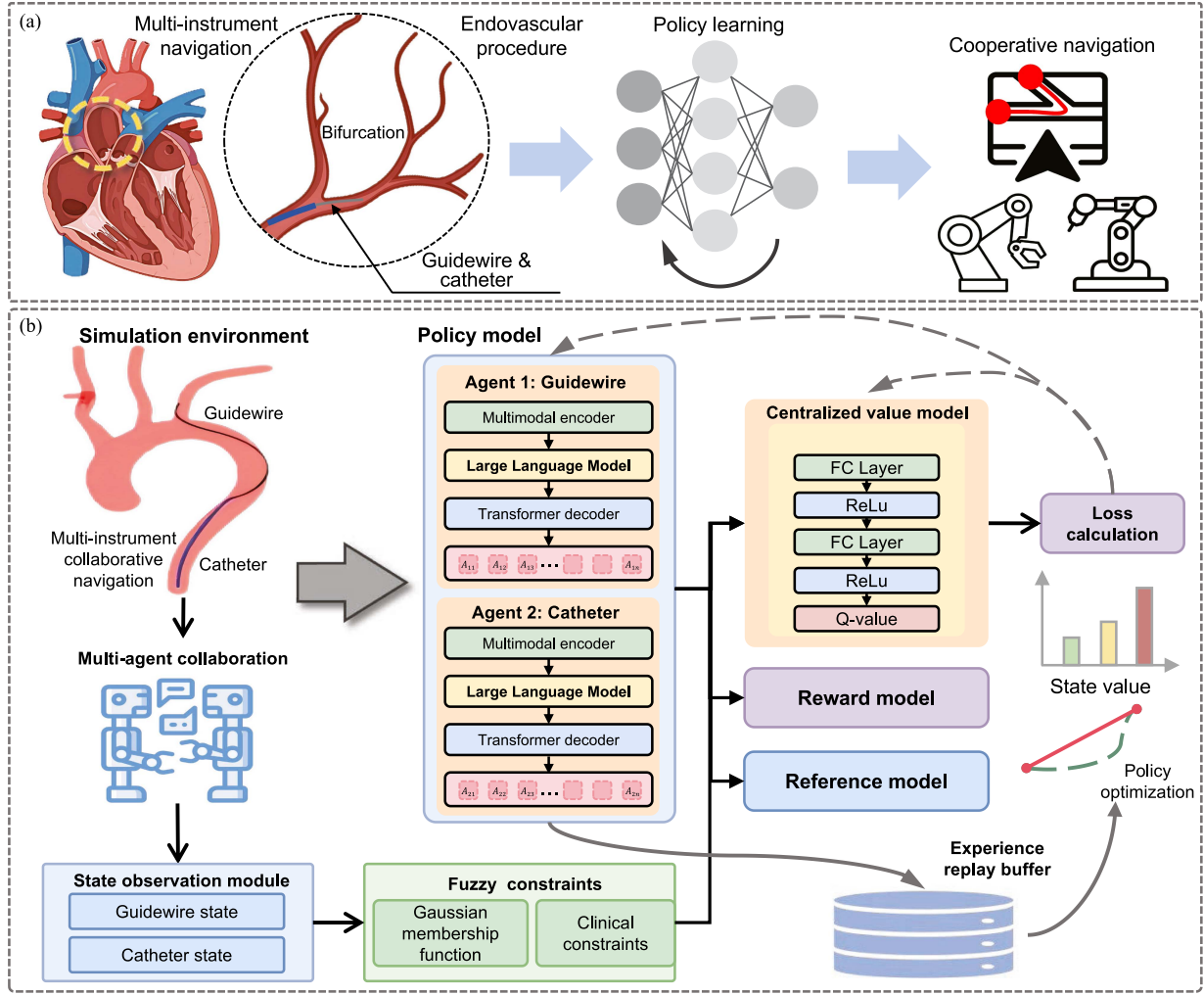


Fig. 3. Schematic overview of the proposed LLM-guided FRL framework for robotic-assisted endovascular navigation. (a) Clinical context and conceptual pipeline: multi-instrument navigation involving a guidewire and catheter within complex vascular structures, facilitated by FRL for cooperative robotic navigation. (b) System architecture of the proposed framework. A simulation environment models multiagent collaboration for guidewire and catheter control. Each agent integrates a multimodal encoder and an LLM with a Transformer decoder for context-aware action generation. The policy model operates under fuzzy constraints derived from Gaussian membership functions and clinical safety rules to address uncertainties. A centralized value model computes Q-values through fully connected layers with rectified linear unit (ReLU) activations, enabling loss calculation and policy optimization. The state observation module captures real-time guidewire and catheter states, supporting dynamic feedback. FRL refines learning via a fuzzy reward model, reference trajectories, and an experience replay buffer to enhance training efficiency and clinical applicability.

reward functions using Gaussian membership functions

$$\mu_{A_j}(x_j) = \exp\left(-\frac{(x_j - c_j)^2}{2\sigma_j^2}\right)$$

where c_j and σ_j encode clinical thresholds, introducing soft penalties (e.g., a gradual reward reduction when the guidewire and catheter are slightly too close, prompting gentle repositioning) or rewards to ensure protocol-compliant coordination. The parameters of the Gaussian membership functions, specifically the centers c_j and widths σ_j , are derived from a combination of clinical heuristics and empirical tuning. Clinical heuristics, informed by expert knowledge from interventional cardiologists, define the initial ranges for key variables including the relative distance between the guidewire and catheter (d_{wc}) and insertion velocity (v_c), ensuring alignment with safety constraints

like minimum interinstrument distances and safe velocity limits. These heuristics are based on clinical guidelines for endovascular procedures, which prioritize preventing vessel wall damage and maintaining stable instrument coordination. Subsequently, the parameters are empirically refined through iterative simulations in the 3-D vascular environment, optimizing for navigation success and robustness across diverse anatomies. This dual approach ensures that the fuzzy membership functions are both clinically relevant and practically effective, balancing safety and performance under uncertainty. Actor networks compute actions as $a_i = \mu_i(o_i; \theta_i^\mu)$, with parameters updated by maximizing the expected return

$$\begin{aligned} \nabla_{\theta_i^\mu} J_i &= \mathbb{E}_{o_i, a_i \sim \pi_i} \left[\nabla_{\theta_i^\mu} \mu_i(o_i; \theta_i^\mu) \cdot \nabla_{a_i} Q_i(S, a_1, a_2; \theta_i^Q) \right] \\ \theta_i^\mu &\leftarrow \theta_i^\mu + \alpha^\mu \nabla_{\theta_i^\mu} J_i. \end{aligned}$$

The critic minimizes the temporal-difference loss

$$L(\theta_i^Q) = \mathbb{E}_{\mathcal{S}, a_1, a_2, r_i, \mathcal{S}'} \left[\left(Q_i(\mathcal{S}, a_1, a_2; \theta_i^Q) - y_i \right)^2 \right]$$

where

$$y_i = r_i + \gamma Q'_i(\mathcal{S}', \mu'_1(o'_1; \theta_1^{\mu'}), \mu'_2(o'_2; \theta_2^{\mu'}); \theta_i^Q)$$

and Q'_i and μ'_i denote target networks with soft updates. LLMs enhance policy networks by generating contextual embeddings, but their outputs exhibit uncertainties due to variability in anatomical data, procedural contexts, and token generation inconsistencies.

To ensure robust and context-aware policy generation, LLM prompts are constructed as structured inputs that combine real-time sensory data and procedural metadata. These prompts integrate critical environmental features, including vascular topology (vessel curvature, bifurcation complexity) and instrument states (including position and velocity), with clinical objectives (target lesion navigation, adherence to safety constraints). The prompt structure is designed to be flexible, dynamically adjusting to the specific vascular anatomy (e.g., Models I–III, varying in tortuosity and branching) and procedural context (e.g., Tasks A–I, differing in navigation complexity). This adaptability is achieved by modulating the emphasis on relevant features, such as prioritizing vessel curvature in tortuous anatomies or instrument coordination in complex tasks, ensuring that LLM-generated policies align with patient-specific and task-specific requirements. The fuzzy system further refines these outputs by applying Gaussian membership functions tuned to specific clinical constraints, enhancing robustness against uncertainties in LLM-generated tokens.

FRL addresses these by embedding T-S fuzzy systems to refine LLM outputs. Each agent's policy module includes a multimodal encoder, an LLM embedding layer, and a transformer-based action decoder [see Fig. 3(b)]. The LLM processes anatomical and procedural data, producing token sequences $T = \{t_1, \dots, t_k\}$. A fuzzy aggregation layer processes these tokens using Gaussian membership functions

$$\mu_{A_j}(t_j) = \exp \left(-\frac{(t_j - c_j)^2}{2\sigma_j^2} \right) \quad (1)$$

where the centers c_j and widths σ_j , as described above, are calibrated through clinical heuristics and empirical tuning to reflect the specific vascular anatomy and procedural requirements. This calibration ensures that LLM outputs are aligned with clinical safety constraints, such as maintaining safe interinstrument distances or limiting insertion velocities.

Mitigating uncertainties in semantic interpretation. The transformer decoder maps fused inputs to actions

$$a_i = \text{Transformer Decoder}(\text{Fuzzy Aggregate}(T, o_i); \theta_i^\pi) \quad (2)$$

enabling robust navigation through vascular bifurcations and curvature transitions (e.g., navigating a sharp bend in a vessel, requiring smooth adjustments to avoid stressing the vessel wall). Vascular topology and instrument positions are fused with LLM

tokens, with fuzzy systems enhancing generalization to patient-specific anatomies.

The T-S fuzzy system, with R rules, processes input vector $\mathbf{x} \in \mathbb{R}^n$, including LLM-derived features, producing output

$$y(\mathbf{x}) = \frac{\sum_{i=1}^R w_i(\mathbf{x}) f_i(\mathbf{x})}{\sum_{i=1}^R w_i(\mathbf{x})}$$

where $w_i(\mathbf{x}) = \prod_{j=1}^n \mu_{A_{ij}}(x_j)$, $f_i(\mathbf{x}) = \mathbf{a}_i^\top \mathbf{x} + b_i$, $i = 1, \dots, R$.

The rule-based outputs of the T-S fuzzy system directly influence action selection by providing a weighted contribution to the final action, ensuring safety-critical constraints are enforced. Each rule i is activated based on the firing strength $w_i(\mathbf{x})$, computed as the product of membership degrees $\mu_{A_{ij}}(x_j)$ for input variables d_{wc} and v_c

$$w_i(\mathbf{x}) = \prod_j \mu_{A_{ij}}(x_j) \quad (3)$$

where $\mu_{A_{ij}}(x_j)$ represents the membership degree of the input variable x_j under the fuzzy set A_{ij} . The consequent function

$$f_i(\mathbf{x}) = \mathbf{a}_i^\top \mathbf{x} + b_i \quad (4)$$

maps the inputs to a candidate action component, such as adjustments to guidewire advancement or catheter rotation. The final output $y(\mathbf{x})$ is a normalized weighted sum of these consequents, expressed as

$$y(\mathbf{x}) = \frac{\sum_i w_i(\mathbf{x}) f_i(\mathbf{x})}{\sum_i w_i(\mathbf{x})}. \quad (5)$$

In this formulation, rules with higher firing strengths (e.g., those triggered by unsafe conditions like $d_{wc} < 2$ mm) contribute more significantly, effectively prioritizing actions that maintain clinical safety (e.g., slowing insertion or increasing interinstrument distance). This output is integrated into the transformer decoder's input, modulating the action a_i to align with safe navigation policies. This system encodes clinical constraints, such as relative distance d_{wc} and insertion velocity v_c , ensuring uncertainty-resilient transitions. Stability is ensured via a Lyapunov function

$$V(\mathbf{x}) = \mathbf{x}^\top P \mathbf{x}, \quad P \succ 0 \quad (6)$$

satisfying

$$\dot{V}(\mathbf{x}) = \nabla_{\mathbf{x}} V(\mathbf{x}) \cdot \dot{\mathbf{x}} < 0 \quad \forall \mathbf{x} \neq \mathbf{0} \quad (7)$$

where $\dot{\mathbf{x}} = f(\mathbf{x}, y(\mathbf{x}))$ represents closed-loop dynamics. FRL integrates this stability condition into the reward structure.

The fuzzy reward function for each agent $i \in \{1, 2\}$, defined as

$$\mathcal{R}_i = R_{\text{goal}} + R_{\text{efficiency}} + R_{\text{safety}} + R_{\text{coordination}} \quad (8)$$

comprises four key components, each designed to promote specific clinical behaviors in endovascular interventions. The term R_{goal} rewards successful navigation to the target, such as guiding the wire to a stenosis in a coronary artery, ensuring the primary objective of lesion treatment is met. $R_{\text{efficiency}}$ promotes

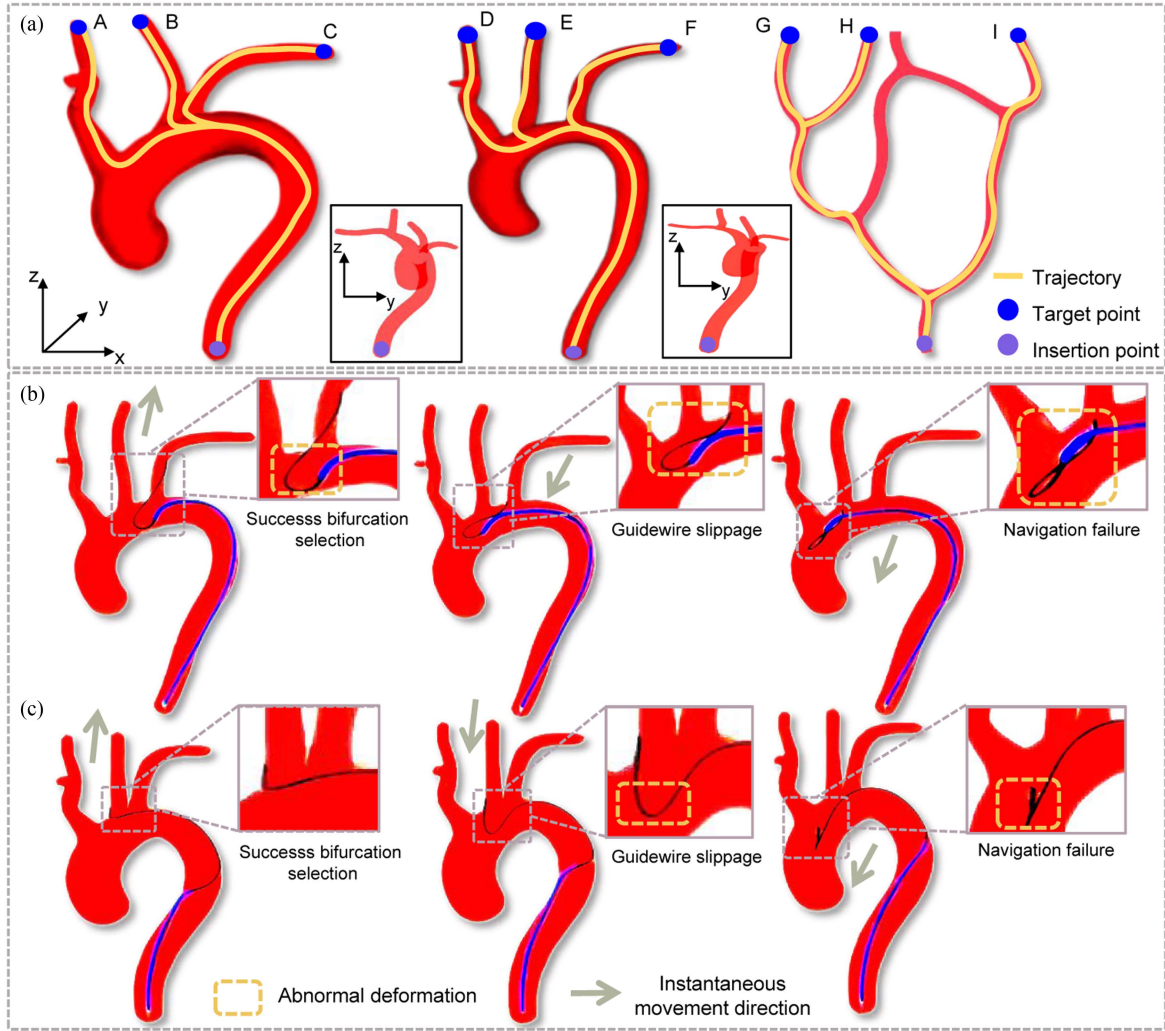


Fig. 4. Experimental analysis of autonomous multi-instrument navigation in simulated vascular environments. (a) Task configuration, showing navigation trajectories from insertion to target in a simulated vascular environment: Vascular Model I (Tasks A–C), Model II (Tasks D–F), and Model III (Tasks G–I). (b) Failure Case 1: illustrates the guidewire and catheter detaching from intended vessel branches due to poor coordination, leading to navigational error. (c) Failure Case 2: shows abnormal deformation, with the guidewire and catheter exhibiting unexpected bending or distortion before entering a bifurcation, accompanied by a sudden, unintended displacement during navigation.

procedural efficiency by penalizing excessive movements or prolonged procedure times, akin to minimizing radiation exposure during fluoroscopy-guided interventions. R_{safety} enforces fuzzy constraints to maintain safe interinstrument distances and limit insertion velocities, preventing vessel perforation or trauma, as might occur during navigation through tortuous vessels. Finally, $R_{\text{coordination}}$ incentivizes synchronized operation between the guidewire and catheter, similar to coordinated movements required to navigate complex vascular bifurcations, ensuring smooth and effective multiagent cooperation.

These components collectively mitigate uncertainties in LLM outputs, target positioning, and instrument coordination, aligning policies with clinical protocols. By integrating fuzzy reward structures with clinical constraints, the FRL framework enables robust, precise, and safe navigation in endovascular robotics, fostering policies that optimize procedural outcomes while adhering to safety and efficiency standards.

IV. EXPERIMENTS

A. Simulation Environment and Computational Framework

The simulation environment utilized the SOFA version 23.06.00 with the Beam Adapter plugin to model deformable object dynamics via beam theory formulations. The RL algorithms were implemented in Python 3.8 using PyTorch 1.11.0, with custom environments created through OpenAI's Gym library to integrate with SOFA. Computations were executed on an NVIDIA GeForce RTX 4070 GPU.

3-D vascular models, utilized in both simulation and physical validation, are derived from CT scans supplied by United Imaging, Ltd., ensuring congruence between virtual and experimental representations. These models are subsequently processed using Blender, a sophisticated 3-D rendering suite, and their coordinate systems are precisely defined with Onshape, a cloud-based computer-aided design platform. Building upon the Sim4EndoR

TABLE I
COMPARISON OF PATH RATIO PERFORMANCE FOR VARIOUS RL METHODS IN ENDOVASCULAR NAVIGATION TASKS ACROSS DIFFERENT VASCULAR MODELS (FAILED CASES)

Method	Vascular Model I			Vascular Model II			Vascular Model III		
	Task A	Task B	Task C	Task D	Task E	Task F	Task G	Task H	Task I
DQN	0.68 ± 0.09	0.64 ± 0.10	0.60 ± 0.08	0.56 ± 0.11	0.52 ± 0.10	0.51 ± 0.09	0.50 ± 0.10	0.52 ± 0.08	0.50 ± 0.09
DDPG	0.73 ± 0.08	0.70 ± 0.07	0.66 ± 0.09	0.62 ± 0.08	0.60 ± 0.10	0.59 ± 0.08	0.58 ± 0.09	0.60 ± 0.07	0.57 ± 0.08
PPO	0.76 ± 0.07	0.72 ± 0.08	0.68 ± 0.06	0.64 ± 0.09	0.62 ± 0.07	0.61 ± 0.08	0.63 ± 0.06	0.60 ± 0.07	0.62 ± 0.08
A3C	0.78 ± 0.06	0.74 ± 0.07	0.70 ± 0.05	0.66 ± 0.08	0.64 ± 0.06	0.63 ± 0.07	0.62 ± 0.06	0.64 ± 0.05	0.61 ± 0.07
MADDPG	0.83 ± 0.05	0.80 ± 0.06	0.77 ± 0.04	0.75 ± 0.05	0.72 ± 0.06	0.71 ± 0.05	0.70 ± 0.06	0.72 ± 0.04	0.69 ± 0.05
LLM (w/o fuzzy)	0.77 ± 0.07	0.74 ± 0.07	0.70 ± 0.06	0.67 ± 0.08	0.65 ± 0.07	0.64 ± 0.07	0.64 ± 0.06	0.65 ± 0.06	0.63 ± 0.07
Ours	0.86 ± 0.04	0.83 ± 0.05	0.80 ± 0.04	0.78 ± 0.05	0.76 ± 0.04	0.74 ± 0.05	0.73 ± 0.04	0.75 ± 0.04	0.72 ± 0.05

framework, the environment is further augmented with stEVE integration, enabling multi-instrument coordination through the incorporation of catheter dynamics. The experimental setup, including the pipeline and task settings for the proposed framework, is illustrated in Fig. 4, where Vascular Models I–III are employed to evaluate navigation tasks A–C, D–F, and G–I, respectively.

B. Implementation Details

This section defines the evaluation parameters for the proposed MAFRL System with LLM for autonomous multi-instrument navigation in robotic-assisted endovascular procedures, ensuring reproducibility and clinical relevance in a simulated environment. The simulation replicates a 3-D vascular network derived from anonymized patient data. The state space $\mathcal{S}$ includes positions, orientations, velocities of the guidewire and catheter, and Euclidean distance to the target lesion. Action spaces $\mathcal{A}_w$ and $\mathcal{A}_c$ permit translational velocity adjustments of ± 0.1 mm/s and rotational angle changes of $\pm 5^\circ$, constrained by clinical safety limits. An environment timestep of 0.02 s balances computational efficiency and realism.

The framework employs LLaMA 2 from the llama-factory toolkit [35] to generate context-aware policies and enable collaborative planning for guidewire and catheter navigation in endovascular interventions. Each agent integrates multimodal encoders and a Transformer decoder to process real-time procedural data and produce policy actions. A centralized value network, consisting of two fully connected layers with 512 and 256 units and ReLU activation, estimates joint Q-values. The system is optimized using the Adam optimizer with learning rates of 1×10^{-5} for the LLM policy model and 5×10^{-4} for the value model, a discount factor of $\gamma = 0.99$, and an experience replay buffer with a capacity of 1×10^6 , sampling mini-batches of 256. Target networks are updated with a soft update parameter $\tau = 0.005$ to ensure stable learning.

Fuzzy constraints are enforced via a T-S fuzzy system to regulate policy outputs and maintain procedural safety. Gaussian membership functions are defined for relative distance between instruments (range [0, 10] mm, centers at 2, 5, 8, width $\sigma = 1.5$) and insertion velocity (range [0, 2] mm/s, centers at 0.5, 1, 1.5, width $\sigma = 0.3$). Five fuzzy rules ensure safety, with stability guaranteed by a positive-definite Lyapunov matrix. The reward function for each agent comprises four components: a goal reward defined as $R_{\text{goal}} = 1800 \cdot (1 - e^{-200 \cdot (t-50)})$ for $t > 50$, an efficiency reward of -5 per episode, a safety reward with

a penalty weight of 300 and threshold of 100, and a coordination reward with a weight of 200 and threshold of 150. These components balance navigation efficiency, safety, and agent collaboration.

Training spans 1000 episodes, each capped at 500 steps, with metrics averaged over 20 runs to ensure statistical robustness. The framework leverages fuzzy logic to mitigate LLM variability, while multiagent coordination optimizes collaborative performance, validated through high navigation success and path efficiency in simulated vascular models.

C. Evaluation Metrics

The proposed framework is evaluated using two key metrics: success rate and path efficiency. Success rate, expressed as $\eta = \frac{N_{\text{success}}}{N_{\text{total}}} \times 100\%$ with $N_{\text{total}} = 20$, captures the percentage of trials where the guidewire and catheter successfully reach the target lesion without breaching clinical constraints, providing a vital gauge of the algorithm's reliability across intricate vascular structures. The path ratio, defined as $\rho = \frac{L_{\text{actual}}}{L_{\text{optimal}}}$, is evaluated in failure cases to assess progress by comparing the actual path length traversed to the optimal geodesic distance along the vessel centerline; a value near 1 suggests near-completion despite the outcome. Averaged over 20 trials, these metrics collectively underscore the framework's robustness and its capacity to pinpoint advancement in unsuccessful attempts, essential for enhancing endovascular navigation precision.

V. RESULTS

A. Task Success Evaluation

The success rates across all tasks are detailed in Table II. The proposed method outperforms most baselines. In Task A, it achieves a 95% success rate, surpassing MADDPG (90%) and others (75%–85%). For Tasks B–I, it records 90%, 90%, 85%, 85%, 75%, 70%, 70%, and 95%, respectively, demonstrating superior precision and safety adherence. This highlights the benefit of integrating fuzzy logic into an LLM-guided multiagent framework, enhancing coordination and constraint compliance. Notably, the table reveals a trend where simpler tasks (e.g., Task A) yield higher success rates across all methods, while more complex tasks (e.g., Task G) show greater variability, with the proposed method maintaining a lead. The smallest margin over MADDPG occurs in Task G (70% versus 65%) and Task H (70% versus 70%), suggesting specific challenges in those scenarios, yet the overall superiority of LLM with Fuzzy Integration is evident.

TABLE II
SUCCESS RATE COMPARISON FOR VARIOUS RL METHODS IN ENDOVASCULAR NAVIGATION TASKS (20 TRIALS)

Method	Vascular Model I			Vascular Model II			Vascular Model III		
	Task A	Task B	Task C	Task D	Task E	Task F	Task G	Task H	Task I
DQN-based Approach	0.75	0.70	0.70	0.65	0.60	0.60	0.65	0.60	0.80
DDPG-based Approach	0.80	0.80	0.75	0.70	0.70	0.65	0.60	0.65	0.80
PPO-based Approach	0.85	0.80	0.80	0.75	0.70	0.65	0.70	0.65	0.85
A3C-based Approach	0.85	0.85	0.80	0.75	0.75	0.70	0.65	0.60	0.85
MADDPG (Multi-Agent)	0.90	0.85	0.85	0.80	0.80	0.70	0.65	0.70	0.95
LLM (w/o fuzzy)	0.80	0.80	0.75	0.70	0.70	0.65	0.65	0.60	0.90
Ours	0.95	0.90	0.90	0.85	0.85	0.75	0.70	0.70	0.95

TABLE III
AVERAGE STEPS TAKEN ACROSS ENDOVASCULAR NAVIGATION TASKS (20 TRIALS)

Method	Vascular Model I			Vascular Model II			Vascular Model III		
	Task A	Task B	Task C	Task D	Task E	Task F	Task G	Task H	Task I
DQN	235.7 $\pm$ 4.8	189.4 $\pm$ 4.6	149.6 $\pm$ 4.4	239.8 $\pm$ 4.9	193.5 $\pm$ 4.7	154.2 $\pm$ 4.5	149.4 $\pm$ 4.4	144.8 $\pm$ 4.3	147.1 $\pm$ 4.4
DDPG	229.3 $\pm$ 4.2	179.8 $\pm$ 4.0	139.7 $\pm$ 3.8	233.4 $\pm$ 4.3	183.9 $\pm$ 4.1	144.5 $\pm$ 3.9	135.7 $\pm$ 3.8	131.2 $\pm$ 3.7	133.5 $\pm$ 3.8
PPO	232.6 $\pm$ 4.5	184.2 $\pm$ 4.3	144.3 $\pm$ 4.1	236.7 $\pm$ 4.6	188.4 $\pm$ 4.4	149.1 $\pm$ 4.2	140.3 $\pm$ 4.1	135.9 $\pm$ 4.0	138.1 $\pm$ 4.1
A3C	238.9 $\pm$ 5.0	192.5 $\pm$ 4.8	154.8 $\pm$ 4.6	240.0 $\pm$ 5.1	196.7 $\pm$ 4.9	159.4 $\pm$ 4.7	155.8 $\pm$ 4.6	151.5 $\pm$ 4.5	153.2 $\pm$ 4.6
MADDPG	219.8 $\pm$ 3.0	169.5 $\pm$ 2.8	129.2 $\pm$ 2.6	224.9 $\pm$ 3.1	174.1 $\pm$ 2.9	134.6 $\pm$ 2.7	124.9 $\pm$ 2.6	120.6 $\pm$ 2.5	123.4 $\pm$ 2.6
LLM (w/o fuzzy)	236.7 $\pm$ 4.9	190.3 $\pm$ 4.2	150.2 $\pm$ 4.6	238.1 $\pm$ 5.1	194.3 $\pm$ 4.7	155.2 $\pm$ 4.3	150.3 $\pm$ 4.9	145.7 $\pm$ 4.6	148.2 $\pm$ 4.2
Ours	214.2 $\pm$ 2.9	164.3 $\pm$ 2.3	124.1 $\pm$ 2.6	219.3 $\pm$ 3.0	169.2 $\pm$ 2.7	129.4 $\pm$ 2.2	119.3 $\pm$ 2.6	115.2 $\pm$ 2.9	118.1 $\pm$ 2.3

B. Endovascular Navigation Efficiency

Navigation efficiency is assessed through path ratios of failed cases (see Table I) and average steps taken (see Table III). In Vascular Model I (Tasks A–C), the proposed LLM-guided with fuzzy enhancement method achieves path ratios of 0.86 ± 0.04 , 0.83 ± 0.05 , and 0.80 ± 0.04 , surpassing MADDPG (0.83 ± 0.05 in Task A) and DQN (0.68 ± 0.09). In Model II (Tasks D–F), it records 0.78 ± 0.05 , 0.76 ± 0.04 , and 0.74 ± 0.05 , outperforming PPO (0.64 ± 0.09 in Task D). In Model III (Tasks G–I), it yields 0.73 ± 0.04 , 0.75 ± 0.04 , and 0.72 ± 0.05 . Table I reveals tighter confidence intervals for the proposed method (± 0.04 in Task H) compared to baselines (± 0.10 for DQN in Task E), demonstrating higher consistency. The method also requires fewer steps (214.2 $\pm$ 2.9 in Task A versus 235.7 $\pm$ 4.8 for DQN), as detailed in Table III. This table indicates a reduction in steps across tasks within each model (from 235.7 to 149.6 for DQN in Model I), with the proposed method recording the lowest values, reflecting more direct navigation paths and reduced complication risks.

C. Learning Convergence Analysis

Fig. 5 presents the success rate trajectories over training episodes. The proposed method demonstrates faster convergence and higher final success rates compared to baseline approaches. This result highlights the effectiveness of integrating FRL with LLMs in improving learning efficiency, which is essential for clinical applications. The system reaches approximately 90% success within 500 episodes across most tasks, while baselines, such as DQN and PPO stabilize at 70% to 80% after 800 episodes. In tasks G to I, which involve higher vascular complexity, all methods converge more slowly. However, the proposed system maintains superior performance with a steeper initial learning curve. This reflects the advantage of FRL in

managing uncertainty and accelerating policy optimization in complex environments.

Analysis of failed cases identifies challenges: guidewire and catheter lodging in complex geometries or deviating into unintended branches. Fig. 4 illustrates two instances where initial bifurcation entry is achieved but subsequent detachment occurs due to inadequate coordination, manifesting as abnormal deformation (excessive bending or distortion) and instantaneous movement (abrupt displacement). These observations underscore the necessity for robust synchronization in navigation tasks. Fig. 4(a) presents the task configuration, depicting navigation trajectories across Vascular Models I–III, with Models II and III featuring sharper bifurcations that elevate failure probability. Fig. 4(b) and (c) detail specific failure modes: in Fig. 4(b), an abrupt shift displaces both the guidewire and catheter, with timestamps marking the critical instant; in Fig. 4(c), the guidewire exhibits excessive bending at a bifurcation, diverging from the proper trajectory. Table I quantifies inefficiencies in failed cases, showing DQN's path ratio of 0.50 ± 0.10 in Task G compared to 0.73 ± 0.04 for the proposed Fuzzy+LLM method, indicating that the latter sustains nearer-to-optimal paths despite unsuccessful outcomes.

VI. DISCUSSION

The proposed MAFRL framework integrates FRL with LLMs to enable cooperative navigation in robot-assisted endovascular interventions. Experimental results demonstrate that MAFRL outperforms conventional RL methods in navigation success, path efficiency, and adherence to clinical constraints, addressing a critical need for real-time multi-instrument coordination between guidewires and catheters in complex vascular environments.

The framework's performance stems from the synergy of LLM-driven anatomical generalization, fuzzy logic's uncertainty management, and multiagent coordination. Unlike

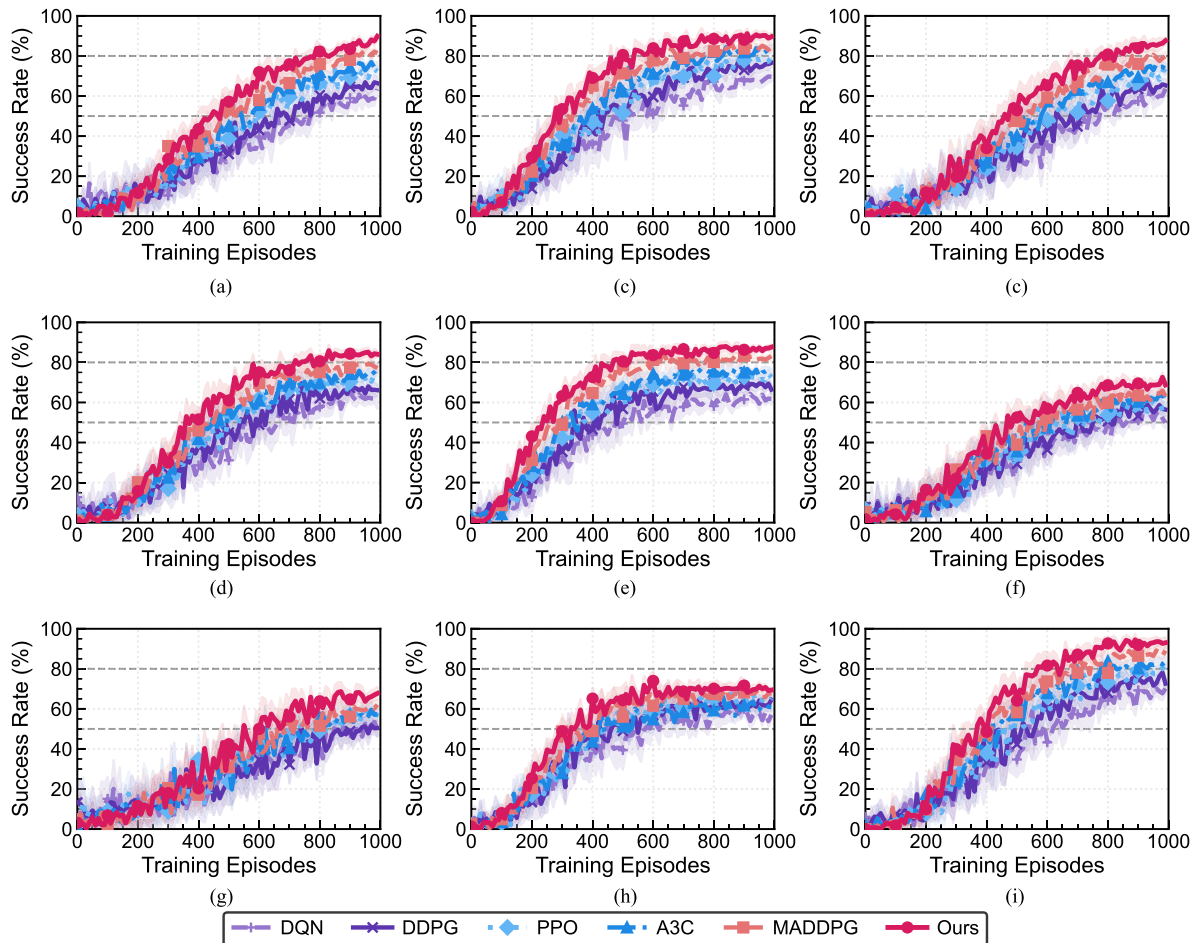


Fig. 5. Success rate curves over training episodes for the proposed method and baseline RL approaches in Task A–I settings. The proposed method (Ours) demonstrates the fastest convergence among all evaluated approaches. (a) Task A. (b) Task B. (c) Task C. (d) Task D. (e) Task E. (f) Task F. (g) Task G. (h) Task H. (i) Task I.

standalone LLM-based navigation, which struggles with uncertainty and occasionally underperforms traditional RL due to ambiguous or noisy states, the fuzzy component embeds clinical heuristics into the reward function and modulates action selection, ensuring robust decision-making. The interpretable fuzzy rule set adapts to diverse interventional styles, adjusting constraints like inter-instrument distance or insertion velocity via operator feedback for safe, personalized navigation. This adaptability also supports training junior interventionalists with real-time guidance, while visualization tools enhance clinician trust and accessibility, advancing clinical applicability.

Despite these strengths, challenges persist in generalizing to rare anatomical anomalies, population variability, or events like vessel spasms, which may require generative model training for robustness across diverse cohorts. Environmental uncertainties, such as sensor noise, image fluctuations, or procedural discontinuities (e.g., imaging pauses), demand robust state retention and further testing. In decentralized execution, complex geometries like tortuous or branched vessels risk agent desynchronization, potentially destabilizing policies; periodic state alignment could mitigate this. Bifurcation misselection, compounded by LLM interpretability challenges, may benefit from fallback logic as a safety net. System-level constraints, including the Transformer's

computational complexity, may introduce latency, necessitating optimizations like model pruning. While Lyapunov-based stability addresses many uncertainties, imaging lag or noise may require extended guarantees.

Clinically, MAFRL promises reduced procedure times and improved patient safety through autonomous, constraint-compliant navigation, with high success rates and low error incidence in simulations. However, validation remains limited to simulated settings, and real-world vascular dynamics, additional instruments beyond guidewire–catheter pairs, and real-time robotic deployment require further exploration. For complex interventions (e.g., stents), enhanced sensory integration is needed, and regulatory validation is critical for translation. The reward structure's parameter sensitivity suggests potential for dynamic adjustments, pending stability analysis. This fusion of LLM reasoning and fuzzy uncertainty management marks a significant step toward intelligent, safe, and adaptive endovascular robotic systems.

VII. CONCLUSION AND FUTURE WORK

This study presents a MAFRL system with LLM for intelligent and cooperative navigation in robot-assisted endovascular

interventions, with demonstrated capability in multi-instrument coordination. By integrating FRL, the framework effectively mitigates the uncertainty limitations of LLM-based navigation, which alone exhibited suboptimal performance in complex clinical environments. The fuzzy module dynamically adapts reward structures and control policies in real time, enabling robust, safety-constrained navigation of multiple instruments, including guidewires and catheters. Experimental results validate that the proposed approach significantly improves navigation success rates, path efficiency, and interinstrument coordination compared to conventional RL methods.

Future work will focus on further expanding the system's clinical applicability by incorporating additional interventional tools, such as balloon catheters and stents, within the cooperative control framework. Deployment on physical robotic platforms with comprehensive in vivo validation will be pursued to ensure clinical reliability. Further refinement of fuzzy rule sets and enhanced utilization of LLM generalization are expected to accelerate policy convergence and strengthen adaptability across diverse anatomical scenarios. These developments aim to translate the demonstrated multi-instrument intelligence into real-world robotic platforms, advancing the precision and intelligence of endovascular interventions.

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